Energy Demand Forecasting Across the New England Grid

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This group project (with Matthew Sheng and Logan Yuan) built a hybrid CNN-Transformer to forecast energy demand 24 hours ahead for each of the 8 ISO New England load zones, using historical and forecasted weather, historical demand, and calendar features as input. Given a weather tensor of shape (450, 449, c), historical demand, and calendar data, the model outputs a sequence of 24 demand vectors, one per zone, for the next 24 hours. All models were scored with MAPE, averaged across the 24-hour horizon and all zones, on a held-out validation set. The project had three parts: train a baseline CNN-Transformer, improve on it, and pursue an independent study. I was responsible for the independent study in Part 3: error analysis on our optimal models and architecture analysis on our final model.

Training Baseline CNN-Transformer

The baseline (Logan's part) downsamples each weather frame with a CNN into a 10×10 grid of spatial tokens, embeds those alongside historical demand, then combines 24 past and 24 future timesteps with learned spatial and temporal position embeddings before a 3-layer, 4-head Transformer encoder predicts the 24-hour demand sequence. Transformer memory scales quadratically, so despite downsampling we were still batch-size-limited to 2. After roughly 10 hours of training, the baseline reached 7% MAPE on the course evaluation script.

Beating the Baseline

Matthew traced how the architecture evolved from the baseline through v2, v3, and v4:

- **Baseline → v2 (the biggest jump):** switched from predicting total load directly to predicting a gated residual on top of a previous-day baseline. This alone cut MAPE from 6.57% to 3.79%, a 42% relative improvement, since the model no longer had to relearn the obvious daily pattern from scratch.
- **v2 → v3 (a negative result):** added later-horizon weighting, ramp-hour emphasis, calendar harmonics, and zone adapters. Despite being reasonable hypotheses individually, v3 (3.98% MAPE) did not consistently beat v2, winning on only 39.5% of validation windows. Ablations of the individual v3 modules all landed within a narrow 3.39%-3.48% MAPE band, so no single idea stood out as reliably better, and were not carried forward.
- **v3 → v4 (the final model):** built a seasonal anchor that supports daily, weekly, and biweekly lags (our final 72-hour history run only used daily lag), switched to a multiplicative residual (better matched to a percentage-error metric like MAPE), and added a system-total-plus-zone-share prediction path for cross-zone consistency. This reached 3.31% MAPE, a 49.6% improvement over baseline, and improved every one of the 8 zones relative to v2.

The main takeaway: the biggest gains came from changing what the model was asked to predict (a correction to a strong seasonal anchor, not the raw load), not from adding more modules.

Independent Study: Extreme Weather Analysis

I wanted to understand where our models actually broke down, so I focused on extreme weather using our validation set (2023). For this analysis, I used the best v2 checkpoint (3.46% validation MAPE), an earlier run than the 3.79% retrain above. For every input window, I aggregated each channel spatially (max over the grid, then averaged over the 24 forecast hours) and flagged a window as extreme if two or more channels fell outside their 5th-95th percentile range, calculated independently per channel. I did not separate extreme heat from extreme cold, which is a limitation worth noting.

On model_v2, MAPE was 3.29% on non-extreme windows versus 3.74% on extreme windows, a 13.7% relative gap. Since that number is averaged across all 8 zones, I broke it down further. Zone-level bias showed the model consistently over-predicted in CT, ME, and VT and under-predicted in NEMA_BOST (Boston) and SEMA (its suburbs) in both conditions. This is possibly from the model initializing all zones with equal attention over a 4×4 spatial tile, so population-dense zones end up dominating the learned weather signal, and Boston's hotter-than-usual 2023 summer [1] likely compounded the under-prediction there. I also looked at which channel combinations drove the worst extreme-weather errors:

- Humid, low-wind, low-solar conditions (a still, cloudy day where neither heating nor cooling dominates) produced the single worst MAPE.
- Hot, humid, sunny combinations were close behind, plausibly tied to heavy AC load.

Both patterns point to the same root cause: the model collapses each channel to one spatially-averaged value before reweighting them, so rare, spatially localized weather combinations get diluted into weights learned from far more common conditions.

Looking at the forecast horizon, model_v2's MAPE was already 2.5-2.7% one hour out, meaning even with the most recent information the model does not predict demand perfectly, and the gap between extreme and non-extreme conditions shows up immediately, not just at longer horizons. Error rose quickly through hour 8, then grew more slowly after that, which could be explain by the model's GRU processing the future window sequentially: once errors accumulate over the first 8-10 hours, the Transformer uses those embeddings has less to work with for hours further out. Hour-of-day told a related story: MAPE peaks during the 17:00 UTC (1 PM EST), and that is also where the extreme/non-extreme gap shrinks the most, at times fully overlapping, meaning the model is about equally confused by normal and extreme weather right at peak demand.

Model_v4 was designed in part to address the v2 issues above. It replaced v2's additive correction with a multiplicative one and added a system-total + zone-share path; a new weather summary encoder tracked each channel's mean, std, min, max, and short and daily deltas rather than one flat spatial average; and it shared one CNN across historical and future frames so both lived in the same embedding space. We also added calendar structure, explicitly flagging the 15-19 UTC peak window and adding an hour×day-of-week embedding so the model can separate weekday and weekend peaks. The result: MAPE improved to 3.16% (non-extreme) and 3.60% (extreme). Both models were evaluated on the same 2023 windows with 24 hours of history, but v4 was trained on 72 hours, so the numbers are possibly conservative. Figure 1 showed both regimes improved together. The more interesting finding was in the bias, not the MAPE: v4 learned to systematically over-predict in nearly every zone, in both conditions (Figure 2). That hurt bias in CT, ME, and WCMA specifically, but it improved overall MAPE by compensating more accurately everywhere else, so the model generalized better by consistently erring in one direction rather than by being unbiased on average. At peak hours specifically, the v2 extreme/non-extreme graphs that used to overlap become clearly separated in v4, suggesting v4 actually learned to distinguish extreme from normal conditions at the hardest part of the day, rather than just averaging over both.

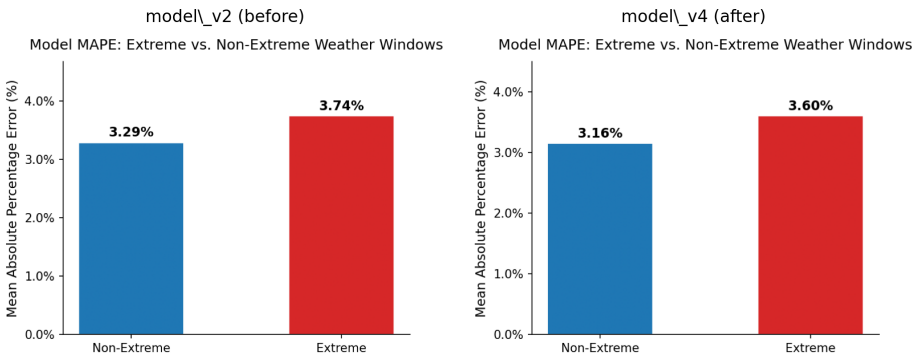


Figure 1: MAPE on extreme vs. non-extreme weather windows, model_v2 vs. model_v4. Both windows improved after the v4 redesign, though the extreme/non-extreme gap persists.

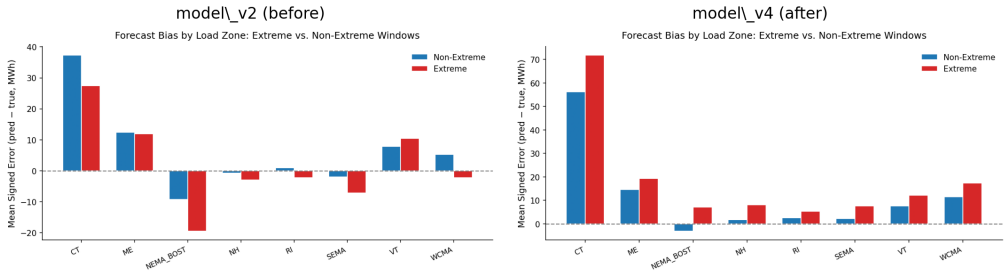


Figure 2: Forecast bias by load zone, model_v2 vs. model_v4. v4 shifted nearly every zone toward over-prediction, worsening bias in CT/ME/WCMA but improving overall MAPE elsewhere.

References

[1] Patrick Whittle and The Associated Press. Sea off New England had one of its hottest years in 2023, part of a worldwide trend. https://www.wbur.org/news/2024/04/29/new-england-ocean-hottest-climate-change, April 2024. WBUR News.